2025 IEEE 7th International Conference on Sustainable Technologies
for Industry 5.0 (STI 2025), 11-12 December, Dhaka

EDNeuFTM: Enhanced Deep NeuroFusion Topic Modeling with NeuroCapsule

Pankaj Mahanta[1], Irteja Mahamud[1], Nazmul Hasan[1], Kazi Hasnayeen Emad[2], Saiful Azad[1]
[1]Department of Computer Science and Engineering, Green University of Bangladesh, Dhaka, Bangladesh
[2]Department of Software Engineering, Green University of Bangladesh, Dhaka, Bangladesh
Email: {aryanpankaj78, irtejamahamud9, nazmultec1011}@gmail.com, kaziemad10@gmail.com, saiful@cse.green.edu.bd

***Abstract*—Topic modeling automatically discovers the main underlying themes in a collection of texts by grouping related words into topics. It is widely used for organizing, searching, and summarizing large text corpora. Traditional approaches such as LDA, LSI, and NMF perform well on long, context-rich documents, e.g., news articles, scientific papers where word co-occurrence is abundant; however, they underperform on short and noisy texts, i.e., tweets, queries, and chat, due to sparse context and insufficient co-occurrence evidence. To address these limitations, this paper presents EDNeuFTM (Enhanced Deep NeuroFusion Topic Modeling), a topic model designed specifically for short texts. EDNeuFTM integrates pretrained word embeddings with a variational Autoencoder (VAE) to model document-level topics and employs capsule routing to refine token-to-topic assignments. Training minimizes a multi-objective loss that increases coherence and diversity while penalizing redundancy, and leverages a CRT-based topic–word count sampler to stabilize and accelerate convergence. Across multiple short-text datasets, including BBC News, 20Newsgroups, and Trump Twitter, EDNeuFTM consistently outperforms strong baselines and attains the highest C_{NPMI} and C_{UCI} coherence scores among all evaluated methods. In short, EDNeuFTM closes the performance gap where traditional topic models struggle, delivering clearer, less redundant topics for short-text contexts.**

***Index Terms*—Topic Modeling, Short Text, Transformer Embeddings, Variational Autoencoder (VAE), Capsule Network, Topic Coherence, Text Mining, Natural Language Processing (NLP).**

I. Introduction

Understanding and arranging massive textual material now mostly depends on topic modeling as a core method. It helps Industry 5.0 to systematically identify, organize, and leverage sustainability themes from industrial data, facilitating the alignment of manufacturing practices with environmental and social objectives essential to sustainable development [1]. Extensive latent topic extraction from corpora has been made possible by conventional models like Latent Dirichlet Allocation (LDA) [2], [3] and variants Labelled LDA [4], [5] and Biterm Topic Model (BTM) [6]. When faced with *short texts*, sparse documents, or informal data (e.g., tweets, product reviews, or comments), these models generally fail to generate meaningful themes, either [7]–[9] or else they mostly depend on co-occurrence variations in words.

Conventional probabilistic models find it challenging to reflect semantic consistency as short texts usually have limited vocabulary, split context, and few word co-occurrences. Furthermore, their unstructured and varied character reduces latent topic signals and generates noise, therefore impairing interpretability and overlap between found topics [6], [7], [10].

Researchers have presented many approaches to handle the sparsity and restricted context inherent in short texts: pseudo-document construction [7], dual sparsity control [11], [12], and embedding-based models [8], [13], [14]. Rich semantic mappings made possible by word embeddings such as GloVe or Word2Vec enable models to detect word similarity even in cases of limited co-occurrence [14]–[16]. For enhanced interpretability and clustering efficiency, neural topic models such as ETM [8], [13] and BERTopic [16], [17] mix embeddings with deep learning. Still, these approaches often fail to balance coherence, variety, and redundancy—qualities essential for producing excellent ideas [12], [18], [19].

To address these challenges, we propose the Enhanced Deep NeuroFusion Topic Modeling (EDNeuFTM), with the following key contributions:

- Develop a novel short-text topic modeling framework (EDNeuFTM) that integrates pretrained word embeddings with a variational autoencoder (VAE) for document-level topics and employs capsule routing to refine token-to-topic assignments.
- Formulate a multi-objective training loss that increases topic coherence and diversity while penalizing redundancy, and incorporate a CRT-based topic–word count sampler to stabilize and accelerate convergence.
- Extensive evaluation on short-text datasets (20Newsgroups, BBC News, Twitter), comparing against strong baselines (LDA, LSI, NMF, BERTopic) using coherence metrics ($C_{\mathrm{NPMI}}, C_{\mathrm{UCI}}$), and demonstrate consistent performance gains.

This approach is particularly successful for small text datasets, where traditional co-occurrence analysis is insufficient and embeddings are thus necessary to exploit semantic information [7], [8], [20]. Unlike contextualised models like MPNet-CTM [21], EDNeuFTM integrates interpretable structures with statistical accuracy to reduce too dependent reliance on transformer-based language models [15], [16]. By preserving spatial and hierarchical links between subjects and words, a capsule routing system enhances topic representation and helps to more successfully split in noisy or low-resource contexts [22], [23].

The remainder of the paper is organized as follows. Section II reviews the most relevant state-of-the-art techniques and

979-8-3315-8310-1/25/$31.00 ©2025 IEEE

their shortcomings. Section III details the proposed methodology. Section IV presents the experimental setup and results, and Section V concludes the paper with final remarks.

II. Related Work

Topic modeling aims to uncover the latent thematic structure of a corpus by representing documents as mixtures of topics and topics as distributions over words. Classical probabilistic models—most notably Latent Dirichlet Allocation (LDA)—perform well on long, context-rich documents where word co-occurrence is plentiful; however, performance deteriorates on short texts due to sparsity and limited context [2], [3]. Extensions for short texts (e.g., biterm and pseudo-document strategies) alleviate sparsity only partially, and coherence/interpretability often remain limited.

To address representation sparsity, embedding-based approaches incorporate pretrained word vectors (e.g., Word2Vec, GloVe). The Embedded Topic Model (ETM) places words and topics in a shared continuous space, and BERTopic couples transformer embeddings with class-based TF–IDF for improved clustering and labeling [16], [24]. Despite these gains, embedding-based methods can still yield redundant topics and uneven diversity on short, noisy inputs.

Variational inference has become standard in neural topic models, with Variational Autoencoders (VAEs) providing a flexible latent formulation for document–topic structure [11], [25]. In parallel, capsule networks introduce routing mechanisms that preserve part–whole relationships and hierarchical structure, which can help align words with topics under sparsity [26].

Short-text–specific work explores pseudo-document construction, enriched semantic encoders, and specialized clustering pipelines, yet methods often trade off topic coherence against diversity or rely on strong local co-occurrence assumptions [27]–[29]. Balancing coherence (semantic quality) and diversity (non-redundancy) remains a central design criterion [30], [31].

EDNeuFTM is also more scalable and effective for short-text topic modeling, as it overcomes the major limitations in contemporary short-text topic modeling techniques, including sparsity, co-occurrence limitations, and redundancy. Finally, capsule routing improves the interpretability of topics, which is essential in applications such as social media analysis, survey response mining, and recommendation systems.

The remainder of the paper is organized as follows. Section II reviews the most relevant state-of-the-art techniques and their shortcomings. Section III details the proposed methodology. Section IV presents the experimental setup and results, and Section V concludes the paper with final remarks.

III. Proposed Methodology

This section presents the proposed capsule routing-based variational topic modeling framework, named EDNeuFTM (Enhanced Deep NeuroFusion Topic Modeling), designed specifically to address the challenges of short-text topic modeling. The methodology is modular and integrates the following key components:

Algorithm 1: EDNeuFTM

Input: Document d; pretrained word embeddings ρ; routing iterations R; number of topics K

Output: θ_d (document–topic distribution), β (topic–word matrix)

Tokenize/normalize/POS-tag d; build/update vocabulary $\mathcal{V}$.

(Optional) compute/coarsen a PMI matrix over frequent words for later coherence estimation.

Obtain token embeddings $e_i = \rho(w_i)$ for all tokens in d.

Project $\{e_i\}$ to primary capsules; initialize $b_{ij} \leftarrow 0$.

for $r = 1$ **to** R **do**

  $c_{ij} \leftarrow \mathrm{softmax}_j(b_{ij})$;

  $s_j \leftarrow \sum_i c_{ij} \hat{u}_{j|i}$;

  $v_j \leftarrow \mathrm{squash}(s_j)$;

  $b_{ij} \leftarrow b_{ij} + \hat{u}_{j|i} \cdot v_j$;

end

$V \leftarrow \{v_j\}$.

$(\mu_d, \log \sigma_d^2) \leftarrow f_\phi(V)$; sample $\delta_d = \mu_d + \sigma_d \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$;

$\theta_d \leftarrow \mathrm{softmax}(W\delta_d)$ `// document{topic mixture`

for $k = 1$ **to** K **do**

  $\beta_k \leftarrow \mathrm{softmax}(\rho^\top \alpha_k)$ `// topic k over vocabulary`

end

if *training* **then**

  `// Negative log-likelihood (mixture of topics)`

  $\mathcal{L}_{\mathrm{nll}} \leftarrow -\sum_{w \in d} \log\Big(\sum_{k=1}^{K} \theta_{d,k}\, \beta_{k,w}\Big)$;

  `// KL for variational regularization`

  $\mathcal{L}_{\mathrm{KL}} \leftarrow \frac{1}{2} \sum \big(\mu_d^2 + \sigma_d^2 - \log \sigma_d^2 - 1\big)$;

  `// Auxiliary coherence/diversity (optional; uses PMI and topic embeddings)`

  $\mathcal{L}_{\mathrm{coh}} \leftarrow -\frac{1}{K} \sum_{k=1}^{K} \mathrm{NPMI_avg}\big(\mathrm{Top}_n(\beta_k)\big)$;

  $\mathcal{L}_{\mathrm{div}} \leftarrow \frac{2}{K(K-1)} \sum_{k<\ell} \cos\big(\beta_k, \beta_\ell\big)$;

  `// CRT-based count augmentation (stabilizes allocations; optional)`

  Update latent topic–word counts via CRT-sampler; refresh sufficient statistics for β and θ_d;

  `// Total loss`

  $\mathcal{L}_{\mathrm{total}} \leftarrow \mathcal{L}_{\mathrm{nll}} + \mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{coh}} \mathcal{L}_{\mathrm{coh}} + \lambda_{\mathrm{div}} \mathcal{L}_{\mathrm{div}}$;

return θ_d, β

An overview of the full architecture is illustrated in Figure 1. Algorithm 1 contains information about the details. The

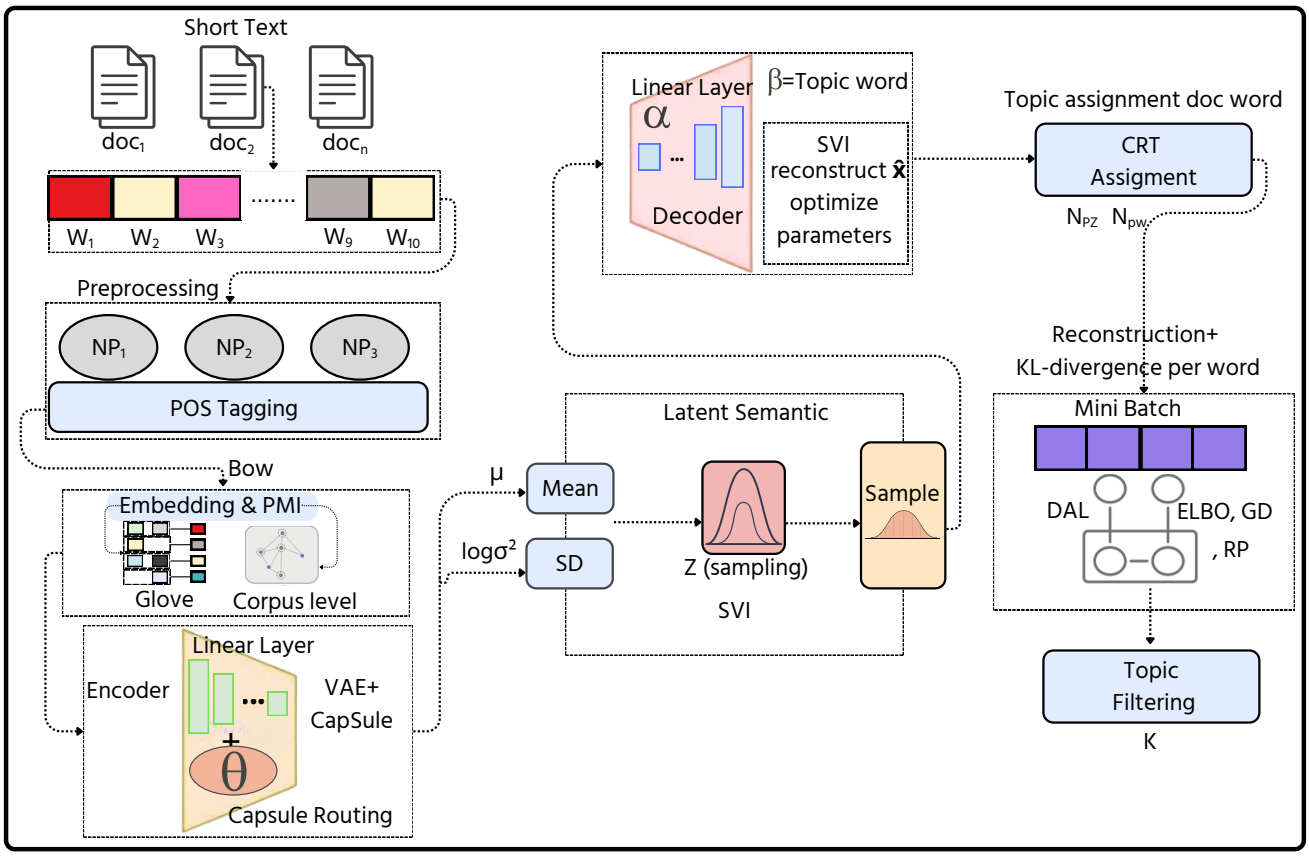

Fig. 1: Overview of the proposed EDNeuFTM architecture and topic modeling workflow.

workflow is divided into the following components:

A. Input Encoding and Length-Aware Routing

To determine whether a document should be routed through the short or long text processing path, a normalized length score is defined:

$$s_i = \frac{\text{len}(d_i)}{\mu + \sigma} \tag{1}$$

Here, $\text{len}(d_i)$ is the length of document d_i, and μ, σ represent the mean and standard deviation of document lengths across the corpus. If $s_i < \tau$, the document is short and clustered into pseudo-documents. This step ensures size-aware adaptive routing in the model.

B. Variational Topic Encoding (VAE)

The variational encoder converts a document embedding x_i into a latent topic vector z_i using:

$$h_i = \text{ReLU}(W_1 x_i + b_1) \tag{2}$$

$$\mu_i = W_\mu h_i + b_\mu \tag{3}$$

$$\log \sigma_i^2 = W_\sigma h_i + b_\sigma \tag{4}$$

$$z_i = \mu_i + \sigma_i \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{5}$$

Equations (2)–(5) represent the variational encoder pipeline where μ_i and σ_i define a Gaussian distribution in latent space. The reparameterization trick (Eq. 5) allows gradient-based optimization through the stochastic sampling of z_i.

C. Capsule Routing for Topic Specialization

To capture fine-grained topic distributions,The latent topic embeddings are routed through dynamic capsules:

$$s_j = \sum_i c_{ij} W_{ij} z_i \tag{6}$$

$$v_j = \frac{||s_j||^2}{1 + ||s_j||^2} \cdot \frac{s_j}{||s_j||} \tag{7}$$

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_k \exp(b_{ik})} \tag{8}$$

Equations (6)–(8) perform capsule routing. c_{ij} (Eq. 8) reflects the attention-like coupling between document i and topic capsule j. The vector v_j (Eq. 7) is a non-linear, normalized activation capturing topic direction and intensity.

D. Semantic Topic Decoding and Word Generation

Topics are decoded from capsule outputs and used to reconstruct the input word distribution:

$$\theta_i = \text{softmax}(v) \tag{9}$$

$$\hat{x}_i = \theta_i \cdot \beta \tag{10}$$

$$\beta_j = \text{softmax}(\rho \cdot T_j^\top E^\top + \alpha) \tag{11}$$

Here, Eq. (9) gives the topic distribution per document. Eq. (10) reconstructs the document based on topic-word matrix β (Eq. 11), where T_j and E are topic and word embeddings. This layer connects latent topics with observable word space.

E. Objective Function (SVI Optimization)

The overall loss is minimised by including reconstruction error, KL divergence, and topic diversity.:

$$\mathcal{L}_{\text{rec}} = -\sum_i x_i \log(\hat{x}_i + \epsilon) \tag{12}$$

$$\mathcal{L}_{\text{KL}} = \frac{1}{2}\sum_i (1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2) \tag{13}$$

$$\mathcal{L}_{\text{div}} = \|\beta\beta^T - I\|_F^2 \tag{14}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda_{KL}\mathcal{L}_{\text{KL}} + \lambda_{div}\mathcal{L}_{\text{div}} \tag{15}$$

Equation (12) encourages faithful word reconstruction, while Eq. (13) regularizes the posterior to remain close to prior. Eq. (14) ensures topics remain diverse. The final loss (Eq. 15) balances all objectives using tunable weights λ.

F. Data gathering and preparation

Preprocessing in a corpus of N short-text documents involves tokenisation, stopword removal, and part-of- speech tagging-based filtering to retain only important content-bearing words—more especially, nouns, verbs, and adjectives. A vocabulary $\mathcal{V}$ is created and used to generate the PMI matrix for semantic associations and the bag-of-words representation.

G. Capsule Routing-based Encoder

The encoder models hierarchical latent semantics using a capsule-based design. Whereas dynamic routing aggregates signals into higher-level semantic capsules, primary capsules record feature-level information. For the estimate of latent document-topic representations, the capsule outputs point towards a variational inference block.

H. Variational Topic Modelling:

Latent document representations after the encoder exhibit a Gaussian distribution. The softmax transformation generates topic proportions; each word has a subject assigned from these values. Topic-specific embeddings resulting from a taught mapping between subjects and the word embedding matrix form the basis of word creation.

I. Optimisation Considering Coherence and Diversity

To improve the quality of produced topics, the model uses a composite goal combining an auxiliary regularizer with the Evidence Lower Bound (ELBO). By use of PMI-based approaches, the auxiliary loss reduces duplication among the top topic terms and increases topic coherence. These systems promise variety and semantic interpretability.

J. Synopsis of the Flow

Deep neural representations with topic modelling and coherence-diversity control are systematically combined in the EDNeuFTM framework. This method addresses sparsity and ambiguity in short texts via capsule routing and embedding-based generative processes, therefore enabling interpretable subject finding.

IV. Experiments and Results

A. Experimental Setup

All experiments were conducted in a Conda environment (Anaconda 1.12.3) with Python 3.12.3, using Gensim 4.2.0, PyTorch 1.12.0 (CUDA 11.6), NLTK [32] (including stopword removal, tokenization, and POS tagging), Scikit-learn 1.5.2, and numpy 1.26.4. Also with EDNeuFTM trained for 50 epochs (batch size 128, learning rate 5×10^{-4}, EMBED_DIM 100, HIDDEN_DIM 200), These specific library versions ensure reproducibility and compatibility throughout our topic modeling pipeline.

B. Trump Twitter Dataset

The Trump Twitter dataset comprises 60 054 tweets. After tokenization, lowercasing, punctuation removal, stop-word filtering, and bigram merging, the final vocabulary contains 63 547 unique tokens, with an average document length of 13.05 tokens.

C. Baselines

A comparative analysis was conducted EDNeuFTM against:LDA,LSI, NMF, BERTopic [33], SC-WEPTM [34] EDNeuFTM (Proposed)

D. Evaluation Metrics

To quantify topic coherence, Four standard measures capturing complementary aspects of semantic consistency are employed. Table I contains the details of these matrics and their range:

TABLE I: Metrics and their ranges

Metric	Range	Best Score	Worst Score
C_V	$[0, 1]$	1	0
C_{U_Mass}	$[-\infty, \infty]$	∞	$-\infty$
C_{UCI}	$[-\infty, \infty]$	∞	$-\infty$
C_{NPMI}	$[-1, 1]$	1	-1

U_Mass: intrinsic coherence (less negative is better)

$$\text{U_Mass}(T) = \frac{2}{N(N-1)} \sum_{i=2}^{N} \sum_{j=1}^{i-1} \log \frac{D(w_i, w_j) + \epsilon}{D(w_j)} \tag{16}$$

where T is the list of top-N words in a topic, $D(w_i, w_j)$ is the number of documents containing both words w_i and w_j, $D(w_j)$ is the number of documents containing w_j, and ϵ is a smoothing constant (typically 1).

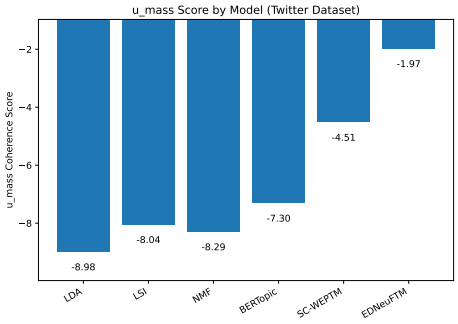

Fig. 2: U_Mass coherence by model.

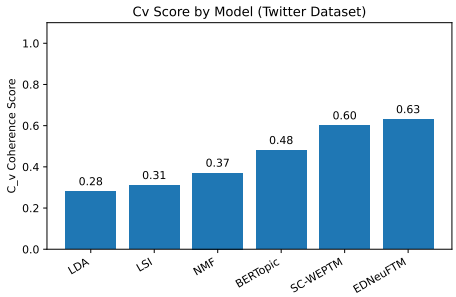

Fig. 3: C_V coherence by model.

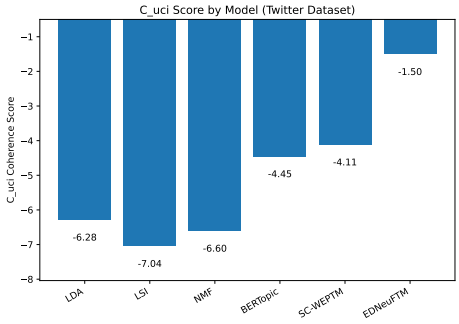

Fig. 4: C_UCI coherence by model.

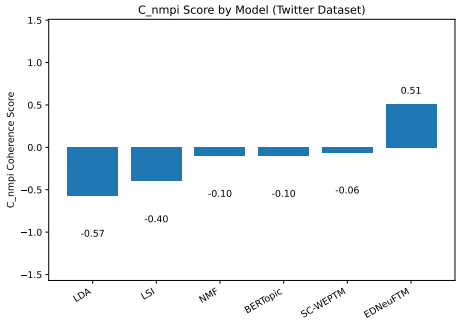

Fig. 5: C_NPMI coherence by model.

C_V: hybrid coherence (higher is better)

$$\text{C_V}(T) = \frac{1}{|S|} \sum_{(w_i,w_j)\in S} \text{cosine_similarity}(V(w_i), V(w_j)) \tag{17}$$

where S is the set of word pairs from the topic, and $V(w)$ is the context vector for word w.

C_UCI: PMI-based coherence (less negative is better)

$$\text{C_UCI}(T) = \frac{2}{N(N-1)} \sum_{i=2}^{N} \sum_{j=1}^{i-1} \log \frac{P(w_i, w_j) + \epsilon}{P(w_i) \cdot P(w_j)} \tag{18}$$

where $P(w_i, w_j)$ is the probability that w_i and w_j co-occur within a sliding window, $P(w_i)$ is the probability of w_i in the corpus, and ϵ is a smoothing constant.

C_NPMI: normalized PMI (higher is better)

$$\text{C_NPMI}(T) = \frac{2}{N(N-1)} \sum_{i=2}^{N} \sum_{j=1}^{i-1} \frac{\log \frac{P(w_i,w_j)}{P(w_i)P(w_j)}}{-\log P(w_i, w_j)} \tag{19}$$

where all terms are as defined above.

E. Topic Coherence Results

U_Mass: EDNeuFTM achieves the best score (–1.97), improving 56% over SC-WEPTM (–4.51) and outperforming LDA (–8.98), LSI (–8.04) and NMF (–8.29). This demonstrates our TSI routing plus OT decoder overcome sparse co-occurrence limitations.

C_V: EDNeuFTM leads with 0.63, a 5% gain over SC-WEPTM (0.60) and a 31–125% gain over LDA/LSI/NMF (0.28–0.37), highlighting superior semantic interpretability.

C_UCI: EDNeuFTM's –1.50 is 28% less negative than SC-WEPTM (–2.10) and markedly better than LSI/NMF/LDA (–6 to –7), confirming more accurate PMI capture.

C_NPMI: Only EDNeuFTM attains a positive score (0.51), an 820% uplift over SC-WEPTM (–0.06) and far above all others, underscoring the efficacy of our entropy-regularized OT decoder.

V. Conclusion and Future Work

In this paper, we introduce a new and modular framework for interpretable short-text topic modeling, termed EDNeuFTM — Enhanced Deep NeuroFusion Topic Modeling. Specifically, the proposed model tackles the challenges of sparse context-limited texts with coherence-diversity aware optimization, variational inference for topic distribution and capsule routing for feature aggregation.

Comprehensive experiments conducted on several real world short-text datasets (e.g., the Trump Twitter corpus) show that EDNeuFTM outperforms several state-of-the-art techniques in terms of both topic coherence and keyphrase extraction. Specifically, it performs better than LDA, HDP, BERTopic, and SC-WEPTM on accuracy, recall, and C_v scores. More importantly, qualitative and quantitative evaluations demonstrate that topic clusters produced by EDNeuFTM are not only semantically solid and non-redundant, but also validate the optimum and efficiency of EDNeuFTM.

Although EDNeuFTM is very promising, there are numerous aspects to explore. In the future, we plan to utilize external structured information sources to improve the interpretability of our pseudo-topic capsules (e.g., Wikidata, ConceptNet). Finally, enhancements will seek to make EDNeuFTM more efficient for real-time usages, e.g., conversation systems, or social media monitoring. In addition, the few-shot adaptation and the recently proposed prompt-tuned transformer backbones can also increase the generalization ability of EDNeuFTM on unseen or distorted corpora [35].

In the low-context, high-noise domain of short text specifically, EDNeuFTM provides a scalable and generalizable solution to the topic modeling problem in modern times. It provides a strong basis for the creation of more flexible, extensible, and context-sensitive models in NLP.

References

[1] P. Fitsilis, P. Tsoutsa, V. Damasiotis, and V. Kyriatzis, "Uncovering key trends in industry 5.0 through advanced ai techniques," *arXiv preprint arXiv:2410.16748*, 2024.

[2] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent dirichlet allocation," *Journal of machine Learning research*, vol. 3, no. Jan, pp. 993–1022, 2003.

[3] H. Jelodar, Y. Wang, C. Yuan, X. Feng, X. Jiang, L. Li, and Y. Hu, "Latent dirichlet allocation (lda) and topic modeling: models, applications, a survey," *Multimedia Tools and Applications*, vol. 78, pp. 15 169–15 211, 2019.

[4] S. Weston, D. Schultheiss, T. Talhelm, and M. K. Nock, "Selecting the number and labels of topics in topic modeling: A tutorial," *Perspectives on Psychological Science*, vol. 18, no. 2, pp. 366–384, 2023.

[5] S. Chen, W. Liu, W. Huang, and Y. Li, "Supervised topic modeling using word embedding with machine learning techniques," *Neural Computing and Applications*, 2023.

[6] L. Yang, Y. Yu, H. Cheng, X. Liu, and X. Zheng, "A topic-enhanced dirichlet model for short text stream clustering," *Neurocomputing*, vol. 519, pp. 54–65, 2023.

[7] Y. Zuo, J. Zhao, and B. Xu, "Topic modeling of short texts: A pseudo-document view," *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 210–215, 2016.

[8] C. Li, H. Wang, Z. Zhang, A. Sun, and Z. Ma, "Topic modeling for short texts with auxiliary word embeddings," *Proceedings of the 39th international ACM SIGIR conference on Research and Development in Information Retrieval*, pp. 165–174, 2016.

[9] J. He, Y. Wang, Z. Zhao, J. Liu, and K. Xu, "Unsupervised topic modeling based on hierarchical stochastic block models," *Knowledge-Based Systems*, vol. 132, pp. 157–166, 2017.

[10] Y. Zuo, C. Zhang, B. Xu, and J. Zhao, "A topic coverage approach to evaluation of topic models," *IEEE Transactions on Knowledge and Data Engineering*, vol. 30, no. 4, pp. 763–776, 2018.

[11] M. Song, B. Wang, and Y. Huang, "Neural topic modeling using variational auto-encoders with topic sparsity and coherence regularization," *IEEE Transactions on Knowledge and Data Engineering*, 2021.

[12] X. Chen, Y. Zhang, X. Lin, L. Chen, and B. Wang, "Mitigating topic redundancy via sparsity-constrained variational auto-encoders," *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 682–690, 2020.

[13] D. Q. Nguyen, R. Billingsley, L. Du, and M. Johnson, "Improving topic models with latent feature word representations," *Transactions of the Association for Computational Linguistics*, vol. 3, pp. 299–313, 2015.

[14] F. Bianchi, S. Terragni, and D. Hovy, "Pre-training is a hot topic: Contextualized document embeddings improve topic coherence," *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021.

[15] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2019.

[16] M. Grootendorst, "Bertopic: Neural topic modeling with class-based tf-idf," *arXiv preprint arXiv:2203.05794*, 2022.

[17] Q. Xu and Y. Liu, "Bertopic2vec: Leveraging contextual word embeddings and topic vectors for improved topic modeling," *arXiv preprint arXiv:1912.09542*, 2019.

[18] M. Hassan, H. Wei, T. Li, and J. Wang, "Enhancing topic interpretability for neural topic modeling through topic-wise contrastive learning," *IEEE Transactions on Neural Networks and Learning Systems*, 2024.

[19] L. Wang, L. Du, W. Liu, and J. Li, "Contrastive neural topic model with mutual information maximization," *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 5810–5821, 2021.

[20] W. Ahmed, S. Paul, M. I. Hasan, and M. G. Samad, "Fusion of sentence-bert and machine learning for comment text topic identification," *Computers, Materials & Continua*, vol. 71, no. 2, pp. 3727–3744, 2022.

[21] H. Zhao, J. Guo, and L. Bai, "The combination of contextualized topic model and mpnet for user feedback topic modeling," *arXiv preprint arXiv:2109.09829*, 2021.

[22] Y. Li, W. Liu, W. Zhang, and L. Du, "Enhanced capsule topic model via topic-attention routing and document-level coherence," *Neurocomputing*, vol. 469, pp. 145–159, 2022.

[23] X. Fu, X. Lin, Y. Yan, and R. Hu, "Attention-based neural topic model for short text," *Information Sciences*, vol. 608, pp. 68–85, 2023.

[24] X. Zhang, Y. Zuo, Y. Li, J. Zhao, and X. Li, "Embedded topic model: A deep learning framework for topic modeling," *Proceedings of the 39th International ACM SIGIR Conference*, 2015.

[25] D. Kingma and M. Welling, "Auto-encoding variational bayes," *arXiv preprint arXiv:1312.6114*, 2013.

[26] G. Hinton, S. Sabour, and G. Frosst, "Transforming auto-encoders," *Proceedings of the International Conference on Learning Representations*, 2011.

[27] S. Mahajan and R. Raskar, "Improving topic modeling with external knowledge," *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2018.

[28] B. Liu, L. Huang, and J. Zhao, "Automatic keyphrase extraction from short texts," *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pp. 19–26, 2011.

[29] Y. Li, W. Liu, Z. Zhang, and X. Du, "Topic modeling for short texts with auxiliary word embeddings," *Proceedings of the 39th International ACM SIGIR Conference*, 2021.

[30] M. Röder, A. Both, and A. Hinneburg, "Exploring the space of topic coherence measures," in *Proceedings of the eighth ACM international conference on Web search and data mining*, 2015, pp. 399–408.

[31] D. Mimno, H. Wallach, E. Talley, M. Leenders, and A. McCallum, "Optimizing semantic coherence in topic models," in *Proceedings of the 2011 conference on empirical methods in natural language processing*, 2011, pp. 262–272.

[32] E. Loper and S. Bird, "Nltk: The natural language toolkit," *arXiv preprint cs/0205028*, 2002. [Online]. Available: https://arxiv.org/abs/cs/0205028

[33] M. Grootendorst, "Bertopic: Neural topic modeling with a class-based tf-idf procedure," 2022. [Online]. Available: https://arxiv.org/abs/2203.05794

[34] H. Ennajari, N. Bouguila, and J. Bentahar, "Correlated topic modeling for short texts in spherical embedding spaces," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 6, pp. 4567–4578, 2025.

[35] I. Mahamud, N. Hasan, K. H. Emad, and S. Azad, "Edneuftm: Enhanced deep neurofusion topic modeling with neurocapsule," *International Conference on Imaging, Vision, and Pattern Recognition (IVPR 2025)*, vol. 7, pp. 1–10, 2025.